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Homework 2

prob 1.)

a.)

Objective:

- Prove the formulas.

- $C_H = T\left(\frac{\partial S}{\partial T}\right)_H$

- $C_M = T\left(\frac{\partial S}{\partial T}\right)_M$

- $dE = TdS + HdM$

- $dG = -SdT - MdH$

Maxwell-relations for Gibbs:

- $\left(\frac{\partial S}{\partial Y}\right)_{T,N_j} = \left(\frac{\partial X}{\partial T}\right)_{Y,N_j}$

Using Reichl Statistical physics book:

- X → gen distance
- Y → gen force

- $\left(\frac{\partial S}{\partial N_j}\right)_{T,Y\{N_{i\neq j}\}} = -\left(\frac{\partial \mu_j}{\partial T}\right)_{Y,N_j}$

Maxwell-relations for internal energy:

- $\left(\frac{\partial T}{\partial X}\right)_{S,\{N_j\}} = \left(\frac{\partial Y}{\partial S}\right)_{X,\{N_j\}}$
- $E = E(S, X)$
- $G = G(T, Y)$

Capacity relations for Gibbs:

- $C_{Y,\{N_j\}} = -T\left(\frac{\partial S}{\partial T}\right)_{Y,\{N_j\}} = -T\left(\frac{\partial^2 G}{\partial T^2}\right)_{Y,\{N_j\}}$
- $C_H = -T\left(\frac{\partial^2 G}{\partial T^2}\right)_H$
- $dE = TdS + HdM$
- $dG = -SdT - MdH$
- $d(dG) = d(-SdT) + d(-MdH)$
- $d^2G = -Sd^2T - dTdS - dHdM - Md^2H$

- $C_H = T\left(\frac{\partial S}{\partial T}\right)_{H,M}$, $C_M = T\left(\frac{\partial S}{\partial T}\right)_{H,M}$
- $S = S(T, H, M)$
- $dS = \left(\frac{\partial S}{\partial T}\right)_{H,M} dT + \left(\frac{\partial S}{\partial M}\right)_{H,T} dM$
- $\left(\frac{dS}{dT}\right)_{H,M} = \left(\frac{\partial S}{\partial T}\right)_{H,M} + \left(\frac{\partial S}{\partial M}\right)_{H,T}\left(\frac{dM}{dT}\right)_{H}$
- $\frac{C_H}{T} = \frac{C_M}{T} + \left(\frac{\partial S}{\partial M}\right)_{H,T}\left(\frac{dM}{dT}\right)_{H}$
  - $\left(\frac{\partial S}{\partial M}\right)_{T,H} = \left(\frac{\partial H}{\partial T}\right)_{M,T}$ (Maxwell-Relation)
    - $\left(\frac{\partial x}{\partial y}\right)_z = \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x$
    - $\left(\frac{\partial H}{\partial T}\right)_M = \left(\frac{\partial H}{\partial M}\right)_T \left(\frac{\partial M}{\partial T}\right)_H$, $\frac{1}{\chi_T} = \left(\frac{\partial H}{\partial M}\right)_T$
- $\boxed{C_H - C_M = \frac{T}{\chi_T}\left(\frac{\partial M}{\partial T}\right)_H^2}$

$$M = M(H, T, S)$$

- $\chi_T = \left(\frac{\partial M}{\partial H}\right)_{T,S}$
- $\left(dM\right)_T = \left(\frac{\partial M}{\partial H}\right)_{T,S}\left(dH\right)_T + \left(\frac{\partial M}{\partial S}\right)_{H,T}\left(dS\right)_T$
- $\left(\frac{dM}{dH}\right)_{T,S} = \left(\frac{\partial M}{\partial H}\right)_{T,S} + \left(\frac{\partial M}{\partial S}\right)_{T,H}\left(\frac{\partial S}{\partial H}\right)_T$
- $\chi_T - \chi_S = \left(\frac{\partial M}{\partial S}\right)_{T,H}\left(\frac{\partial S}{\partial H}\right)_T$

$$\left(\frac{\partial M}{\partial S}\right)_{T,H} = \left(\frac{\partial T}{\partial H}\right)_S$$

- $\left(\frac{\partial x}{\partial y}\right)_z = \left(\frac{\partial x}{\partial z}\right)\left(\frac{\partial z}{\partial y}\right)$
- $\left(\frac{\partial T}{\partial H}\right)_S = \left(\frac{\partial T}{\partial S}\right)_H\left(\frac{\partial S}{\partial H}\right)_T$
- $\left(\frac{\partial T}{\partial S}\right) = \frac{1}{\left(\frac{\partial S}{\partial T}\right)_H} = \frac{T}{C_H}$
- $\chi_T - \chi_S = \frac{T}{C_H}\left(\frac{\partial S}{\partial H}\right)_T\left(\frac{\partial M}{\partial T}\right)$

- $\left(\frac{\partial S}{\partial H}\right)_T = \left(\frac{\partial M}{\partial T}\right)_{S,H}$

- $\chi_T - \chi_S = \frac{T}{C_H}\left(\frac{\partial M}{\partial T}\right)_H^2 \Big)_H$

b.) - $C_H - C_M = \frac{T}{\chi_T}\left(\frac{\partial M}{\partial T}\right)_H^2$

- $M = \frac{C}{T}H$, $C > 0$

- $\frac{\partial M}{\partial T} = CH\frac{\partial}{\partial T}\left(\frac{1}{T}\right) = \frac{-CH}{T^2}$

$T^{-1} \rightarrow -1T^{-2}$

- $\left(\frac{\partial M}{\partial T}\right)^2 = \left(\frac{-CH}{T^2}\right)^2 = \frac{C^2H^2}{T^4}$

- $C_H - C_M = \frac{T}{\chi_T}\left(\frac{C^2H^2}{T^4}\right) = \frac{C^2H^2}{\chi_T T^3}$

Prob 2.)

a.) 1-Component system:

- $\phi = E - TS - \mu n$.
- Prove that $\phi = -pV$.

Grand Potential:

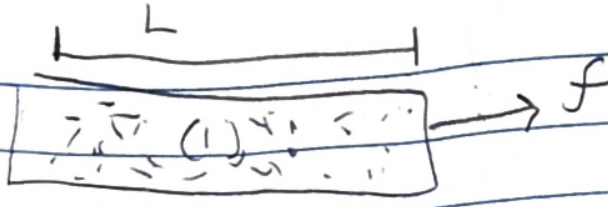
- Thermo potential with T, X, μ_i (gen distance) are fixed
- $E = ST + YX + \mu n$.

(1-component system)

$\phi = \cancel{ST} + YX + \cancel{\mu n} - \cancel{TS} - \cancel{\mu n} = YX$

- $Y = -p$ (gen force), $X = V$ (gen distance/size)
- $\boxed{\phi = XY = -pV}$

b.)



- $\mu \rightarrow$ chemical potential of rubber band
- $n \rightarrow$ number of moles of rubber band
- $TdS \geq dQ = dE - fdl - \mu dn$ (1-component) (1)
- $TS = E - fl - \mu n$ (1-component)
- $d(TS) = d(E) + d(-fl) + d(-\mu n)$
- $TdS + SdT = dE - fdl - ldf - \mu dn - ndM$ (2)
- Subtract (2) and (1).

$$TdS + SdT - dE + fdl + ldf + \mu dn + ndM = 0$$

$$- \quad TdS - dE + fdl + \mu dn$$

- $SdT + ldf + ndM = 0$

$$SdT + \ell d\mathcal{S} + n d\mu = 0$$

C) (PVn system)

1-Component system:

- assume we are working in reversible process.

- $U = ST - PV + \mu n$
- $\frac{U}{n} = \mathcal{U}$ (molar energy)
- $\frac{S}{n} = \mathcal{S}$ (molar entropy)
- $\frac{V}{n} = \mathcal{V}$ (molar volume)
- $\mathcal{U} = \mathcal{S}T - P\mathcal{V} + \mu$
- $T \rightarrow$ temperature.

$\frac{Q}{n} = \mathcal{Q}$ (molar heat)

$\frac{W}{n} = \mathcal{W}$ (specific energy)

- $d(\mathcal{U}) = d(\mathcal{S}T) - d(P\mathcal{V}) + d\mu$

$= d\mathcal{U} = \mathcal{S}dT + Td\mathcal{S} - Pd\mathcal{V} - \mathcal{V}dP + d\mu$

$= d\mu = d\mathcal{U} - \mathcal{S}dT - Td\mathcal{S} + Pd\mathcal{V} + \mathcal{V}dP$ (1)

- $\Delta\mathcal{U} = \Delta\mathcal{Q} + \Delta\mathcal{W} \Rightarrow d\mathcal{U} = d\mathcal{Q} + d\mathcal{W}$

(Reversible process) $\left(d\mathcal{W} = -Pd\mathcal{V},\ d\mathcal{Q} = Td\mathcal{S}\right)$

- $dU = TdS - PdV$ (2)
  - substitute equation (2) into (1).
- $dM = TdS - PdV - SdT - TdS + PdV + VdP$
  - $dM = -SdT + VdP$
    - for differentiating V at const T, $dT = 0$
- $\left(\frac{dM}{dV}\right)_T = V\left(\frac{\partial P}{\partial V}\right)_T$

d.)

Stability theory:

a.)

i.) $\left(\frac{\partial P}{\partial v}\right)_T < 0$

- Stability condition:
  - $-\left(\frac{\partial P}{\partial V}\right)_{T,\{N_i\}} > 0$
  - $\left(\frac{\partial P}{\partial V}\right)_{T,\{N_i\}} < 0$

$V \rightarrow$ is related to specific volume

- $\left(\frac{V}{n} = v\right)$

ii.) $\left(\frac{\partial P}{\partial v}\right)_T < 0$ is true

earlier:

$$\left(\frac{\partial \mu}{\partial V}\right)_T = V\left(\frac{\partial P}{\partial V}\right)_T$$

so $\left(\frac{\partial \mu}{\partial V}\right)_T < 0$

iii.) $\left(\frac{\partial \mu}{\partial V}\right)_T < 0$ is true

b.)

- $\left(\frac{\partial x}{\partial y}\right)_z = \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x$
- $\left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial T}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T$
- $dA = -SdT - pdV - \mu dN$ (Helmholtz)

$$\left(\frac{\partial S}{\partial V}\right)_{T,\{N_i\}} = -\left(\frac{\partial P}{\partial T}\right)_{V,\{N_i\}}$$

- $\left(\frac{\partial T}{\partial S}\right)_V > 0$, and $\left(\frac{\partial S}{\partial V}\right)_{T,\{N_i\}} < 0$

$\left(\frac{\partial T}{\partial V}\right)_S < 0$

$\left(\frac{\partial T}{\partial S}\right)_V$ is true

ii.)

Proved earlier $\left(\frac{\partial P}{\partial T}\right)_V > 0$ then $\left(\frac{\partial T}{\partial t}\right)_s > 0$ is false, So $\left(\frac{\partial P}{\partial T}\right)_V > 0$ is false.

actually $\left(\frac{\partial P}{\partial T}\right)_V < 0$.

e.)

- $A = U - ST =$
  - A → free energy.
  - U → internal energy.
  - T → temperature.
  - S → entropy.

- reversible process:

- $\Delta A = \Delta U - T\Delta S$
- $\Delta U = T\Delta S + P\Delta V$

- $\rightarrow \Delta A = P\Delta V$

- $\rho = \frac{n}{V}, \frac{P}{RT} = \frac{P}{(1-bP)} - \frac{aP^2}{RT}$.

$\rho = \frac{n}{V}$ for any potential to be unstable:

$\left(\frac{\partial^2 U}{\partial X^2}\right)_{X=eq} < 0$ for instability.

$\Delta A \rightarrow$ thermo potential for system.

Find the sign of $\left(\frac{\partial^2 P}{\partial \rho^2}\right)$.

- $P = \frac{pRT}{(1-bp)} - ap^2$
- $\left(\frac{\partial p}{\partial p}\right)_T = RT \frac{\partial}{\partial p}\left(\frac{p}{(1-bp)}\right) - 2ap$
- $\frac{\partial}{\partial p}\left(\frac{p}{(1-bp)}\right) = \frac{(1-bp) - p(-b)}{(1-bp)^2}$

$= \frac{(1-bp) + pb}{(1-bp)^2} = \frac{1}{(1-bp)^2}$

- $\left(\frac{\partial p}{\partial p}\right)_T = \frac{RT}{(1-bp)^2} - 2ap$

$\left(\frac{\partial^2 p}{\partial p^2}\right)_T = RT \frac{d}{dp}\left(\frac{1}{(1-bp)^2}\right) - 2a$

- $\frac{\partial}{\partial p}\left(\frac{1}{(1-bp)^2}\right) = \frac{-(1)2(1-bp)(-b)}{(1-bp)^4}$

$= \frac{2b(1-bp)}{(1-bp)^4} = \frac{2b}{(1-bp)^3}$

$\left(\frac{\partial^2 p}{\partial p^2}\right)_T = \frac{2RTb}{(1-bp)^3} - 2a$

- $\left(p < \frac{1}{b}\right)$
- $\left(b < \frac{1}{p}\right)$

$$(1-bp)^3 = (1-bp)(1-bp)(1-bp)$$

$$= (1-bp-bp+b^2p^2)(1-bp)$$

$$= (1-2bp+b^2p^2)(1-bp)$$

$$= (-bp+2b^2p^2+b^3p^3+1-2bp+b^2p^2)$$

$$= -b^3p^3+3b^2p^2-3bp+1$$

$$\left(\frac{\partial^2 p}{\partial \rho^2}\right)_T = \frac{2RTb}{(-b^3p^3+3b^2p^2-3bp+1)} - 2a > 0$$

$$2RTb > 2a(1-bp)^3$$

$$T > \frac{a}{Rb}(1-bp)^3$$

If $T < \frac{a}{Rb}(1-bp)^3$

unstable